

Vishra Shah

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Professional Summary

Healthcare analytics professional with 2+ years of experience drawing insightful conclusions from health data. Passionate about leveraging data-driven outcomes to improve patient safety, optimize business processes and support evidence-based innovations by increasing continuous improvement in sectors of public health, medical research, health insurance & ecosystem of healthcare start-ups.

Work Experience

Specialist for Claims and Authorization - Premier Oncology Hematology Associates

June 2025 - Present

- Reviewing and analyzing **real-world health insurance claims and authorizations**, gaining hands-on experience in resolving denials, submitting appeals, and supporting redeterminations while maintaining HIPAA compliance.
- Verifying patient insurance eligibility and benefits including co-pays, deductibles, out-of-pocket maximums using payer portals and electronic medical record (EMR) systems.
- Partnered with insurance providers and other stakeholders to address **outstanding claims, reverse denials or delays, and ensure timely posting of insurance payments**, strengthening problem-solving and critical thinking.

Research Data Analyst - Black Men's Health Clinic

Sep 2024 - June 2025

- Standardized and synthesized healthcare data for marginalized communities, improving data accessibility and enabling faster data reporting using **Excel, SQL and Python** for enhanced healthcare delivery for 10 months.
- Created a detail-oriented business plan in a team of 6 to demonstrate viability to potential investors by conducting **market research analysis**.
- Collaborated with stakeholders to draft strategy for data collection & measure program impact to apply for AmeriCorps grant.

Observational Health Data Analyst - Johnson and Johnson

March 2025 - May 2025

- Performed data extraction, transformation, and analysis using **Amazon Redshift** by executing complex SQL queries to analyze **60,000+ records of Real World Data** which included **Medicare Claims data, EMR data, patient reported outcomes, survey data and Iphone and Apple watch sensors data** for early detection of Atrial Fibrillation.
- Performed advanced data analysis in RStudio to identify and resolve data quality issues such as duplicates, missing values, and ETL errors; utilized Quarto to develop an interactive website for presenting analytical findings and insights.
- **Utilized OHDSI tools (ATLAS, OMOP CDM) to map observational health data**, for doing Rural Urban Recruitment Analysis to understand distribution of participants throughout the country.

Research Data Analyst - Franciscan Health Alliance

Oct 2024 - Jan 2025

- Extracted and analyzed data from **Epic** to create a predictive model for 769 patients, utilizing statistical techniques and ML models using **RStudio** to identify key variables associated with extended Length of Stay (LOS) by analyzing 6 months data.
- Conducted **second-pass analysis** for over 1000 Heart Failure patients from Egnite Cardiacare to optimize resource management.
- Analyzed data from 219 Heart Failure patients to compare the proportion of triple and quadruple GDMT before and after pharmacotherapy intervention thus reducing the rehospitalization rate to 7%.

Data Scientist - GTechnologies Pty. Ltd

Jan 2024 - Apr 2024

- Configured and optimized the intelligent EMR system backend development environment using **Excel, SQL and DBeaver**, created 100 tables, to streamline data creation, insertion, and retrieval processes for enhanced efficiency.
- Developed a user-friendly frontend interface utilizing **ReactJS and Node.js** to facilitate seamless integration with the backend system, including formation of forms, headers, and footers components to improve user experience by 5%.
- **Collaborated with clients and end-users** to gather requirements and feedback throughout the **project timeline**, effectively addressing challenges during development.

Teaching Assistant - Indiana University

Aug 2023 - May 2024

- Mentored 136 students in the **Clinical Decision Support Systems** course to increase academic performance by 4%.
- Assisted instructor in incorporating over 20 case studies to optimize learning experiences and facilitate interactive environment.
- Facilitated communication between national and international students, fostering a **collaborative environment & teamwork**.

Research Assistant - Indiana University

Jan 2023 - May 2024

- Developed a **research database** that will be beneficial for academic research, grant proposals, and analytical skill development using Excel, SQL and Python by collecting data within 18 months.
- Conducted data extraction, cleaning and validation by working through 10+ different government healthcare registries.

Operations Associate - Archimedes India Associates

May 2021 - May 2022

- Spearheaded the integration of the Blood Bank IT module into the **IQVIA HIS** reducing manual data entry errors by 10%.
- Digitized patient information through a comprehensive data analysis project, improving patient satisfaction scores by 15%.
- Enhanced patient engagement and medical record access by updating the **Patient Portal**, increasing usage by 15%.

Skills/Knowledge

Languages: SQL,R,Python
Data Standards: HL7, FHIR, SNOMED, ICD-9, ICD-10, LOINC, CPT
EHR and Health IT tools: Epic, RedCap, ATHENA, UpToDate, OHDSI (ATLAS), Symedical, OpenEMR, rHealth, BlueButton, CardioCare, FindHelp, IQVIA EMR.
Insurance Intelligence: Claims Processing, Prior Authorizations, Eligibility Verification (834 files), Denial Management, Appeals & Redeterminations, Coordination of Benefits (COB), EOB Review, Medicare & Medicaid Knowledge, Payer Communication.
Security and Privacy Laws: HIPAA, HITECH, MACRA, MIPS, APMs, CISA
Healthcare Quality Metrics : HEDIS, ACO, MSSP, Star Ratings
Tools and Frameworks: Amazon Redshift, Tableau, Power BI, Quarto, MS Excel, MS Word, MS PowerPoint, Qualtrics, EndNote, RapidMiner, Bitbucket, Jupyter, RStudio, Visual Studio Code, phpMyAdmin, SQLite, MySQL, PostgreSQL, DBeaver.

Education

Indiana University Purdue University, Indianapolis <i>Master of Science, Health Informatics</i>	<i>Indianapolis, USA</i>
National Forensic Sciences University <i>Master of Business Administration, Hospital and Healthcare Management</i>	<i>Gandhinagar, India</i>
Gujarat University <i>Bachelor of Dental Surgery</i>	<i>Ahmedabad, India</i>

Projects

Clinical Decision Support System (CDSS) Prototype Proposal ◦ Created a prototype proposal in a team of 5 to trigger personalized alerts in OPENEMR for COPD patients having smoking history for ensuring timely preventive interventions using SYMEDICAL, OHDSI and UpToDate . ◦ Conducted more than 20 extensive literature reviews to identify relevant research studies and data. ◦ Integrated OPENEMR data and ATLAS cohort information to build a comprehensive decision tree and streamlined Clinical Decision Rules (CDR) enabling more precise and personalized care recommendations to physicians.	<i>May 2023 - Aug 2023</i>
Unveiling Lung Cancer Predictors ◦ Conducted statistical analysis on a dataset of 1,000 lung cancer cases, employing Fisher’s Exact Test and Logistic Regression to identify passive smoking as the highest-risk lifestyle factor for lung cancer development. ◦ Utilized R and data visualization techniques to analyze the correlation between lifestyle variables, such as obesity and smoking, and lung cancer levels, presenting findings through histograms and relationship graphs. ◦ Developed and validated predictive models, optimizing AIC and deviance metrics, to support public health initiatives in understanding the impact of lifestyle choices on lung cancer.	<i>Jan 2023 - May 2023</i>
ML - Powered Predictive BioInformatics Study ◦ Developed and implemented machine learning models , including Random Forest and K-Nearest Neighbors, achieving an 82% accuracy rate for predicting optimal mushroom growth seasons using imbalanced datasets by cross collaborating in a team of 7. ◦ Conducted exploratory data analysis and feature engineering using Python and SQL , improving data quality by addressing 20% null values and removing duplicates across 60,000+ records. ◦ Optimized model performance through data resampling techniques such as SMOTE, enhancing prediction reliability for minority class data in a multiclass classification task.	<i>Aug 2022 - Dec 2022</i>
Project Management Research Project ◦ Proposed and conceptualized a smart medication dispenser integrated with a mobile app (“MedEazy”) to enhance medication adherence through automated reminders, voice assistant alerts, and multi-compartment storage. ◦ Developed a comprehensive project plan, including requirements gathering, stakeholder management, and implementation strategies, to simulate real-world healthcare technology deployment. ◦ Simulated end-to-end project lifecycle, incorporating system design, testing protocols, and market analysis, ensuring alignment with healthcare compliance standards.	<i>Aug 2022 - Dec 2022</i>